

Esther Rolf

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Research focus: *statistical and geospatial ML, with applications in environmental monitoring.*

Education

Ph.D. Computer Science , University of California, Berkeley	2016-2022
Advisors: Ben Recht & Michael I. Jordan	
B.S.E. Computer Science and Engineering , Princeton University (<i>Summa cum laude</i>)	2012-2016
Advisors: Mark Braverman (Senior Thesis), Elad Hazan (Junior Research)	

Academic Positions

Assistant Professor , University of Colorado Boulder Computer Science	2024-
Postdoctoral Fellow , Harvard University Data Science Initiative & Harvard Center for Research on Computation and Society	2022-2024
Postdoc mentor: Milind Tambe	

Affiliations

Research Fellow , Sustainability Research Initiative (SRI) at CU Boulder	2026-
Faculty Fellow , SPIKE Center for Sustainability Education at CU Boulder	2026
AI/ML Lead , Environmental Data Science Innovation and Impact Lab (ESIIL)	2025-
Scientific Director , Togo Data Lab	2024-

Honors & Awards

AI Recognition Award: AI for Scholarly & Creative Work University of Colorado System	2026
Sustainable Development Goals Digital Game Changers Award United Nations Development Programme and the International Telecommunication Union	2023
Rising Star in AI for Social Good Harvard Center for Research on Computation and Society	2021
Best Paper Award Conference on Neural Information Processing Systems (NeurIPS) Joint Workshop on AI for Social Good	2019
Best Paper Award International Conference on Machine Learning (ICML)	2018

Funding

CU Boulder Research and Innovation Office New Frontiers Launch Grant “New Frontiers in Physically-Grounded Geospatial AI for Understanding Complex Global Phenomena with Multi-Modal Data”, PI: Esther Rolf, Co-PIs: Steven Nerem, Morteza Karimzadeh, amount: \$200,000 (my amount: \$125,192)	2026-2027
Taylor Geospatial “Neural Fields of the World”, PI: Esther Rolf, amount: \$399,657 (my amount: \$399,657)	2026-2028
CU Boulder Research and Innovation Office New Frontiers Planning Grant “New Frontiers in Physically-Grounded Geospatial AI for Understanding Complex Global Phenomena with Multi-Modal Data”, PI: Esther Rolf, Co-PIs: Steven Nerem, Morteza Karimzadeh, amount: \$50,000 (my amount: \$50,000)	2025-2026
Patrick J. McGovern Foundation “Cultivating Data Science Capacity in Africa: Piloting a Data Lab in Togo”, PI: Tamma Carleton w/ the Center for Effective Global Action, amount: \$500,000 (my amount: \$65,267)	2024-2025
National Science Foundation DBI Grant 2153040 “Environmental Data Science Innovation and Impact Lab (ESIIL): Accelerating Discovery by Fostering an Open and Diverse Earth Data Revolution”, PI: Jennifer Balch, amount: \$19,900,000 (my amount: \$374,445)	2022-2027
Harvard Data Science Initiative Postdoctoral Fellowship	2022-2024
Harvard Center for Research on Computation and Society Postdoctoral Fellowship	2022-2024
Google PhD Fellowship	2020
National Science Foundation Graduate Research Fellowship (NSF GRFP)	2016
Berkeley Stonebraker Fellowship	2016

Publications

* → Equal contribution

Published Journal Articles

1. Klemmer, K., **Rolf, E.**, Rußwurm, M., Camps-Valls, G., Czerkawski, M., Ermon, S., Francis, A., Jacobs, N., Kerner, H. R., Mackey, L., Mai, G., Mac Aodha, O., Reichstein, M., Robinson, C., Rolnick, D., Shelhamer, E., Sitzmann, V., Tuia, D. & Zhu, X. Earth Embeddings: Towards AI-centric Representations of our Planet. *IEEE Geoscience and Remote Sensing Magazine* (2026).
2. **Rolf, E.**, Gordon, L., Tambe, M. & Davies, A. Contrasting local and global modeling with machine learning and satellite data: A case study estimating tree canopy height in African savannas. *Journal of Machine Learning Research, Special Issue on ML Addressing Problems of Climate Change* (2026).
3. **Rolf*, E.**, Proctor*, J., Carleton*, T., Bolliger*, I., Shankar*, V., Ishihara, M., Recht, B. & Hsiang, S. A generalizable and accessible approach to machine learning with global satellite imagery. *Nature Communications* (2021).
4. Hsiang, S., Allen, D., Annan-Phan, S., Bell, K., Bolliger, I., Chong, T., Druckenmiller, H., Huang, L. Y., Hultgren, A., Krasovich, E., Lau, P. L., Lee, J., **Rolf, E.**, Tseng, J. & Wu, T. The effect of large-scale anti-contagion policies on the COVID-19 pandemic. *Nature* (2020).
5. **Rolf*, E.**, Fridovich-Keil*, D., Simchowitz, M., Recht, B. & Tomlin, C. A successive-elimination approach to adaptive robotic source seeking. *IEEE Transactions on Robotics* (2020).

Published Conference Articles

1. Betti, L., Sanni, F., Sogoyou, G., Agbagla, T., Molitor, C., Carleton, T. & **Rolf, E.** Mapping on a Budget: Optimizing Spatial Data Collection for ML. in *Proceedings of the AAAI Conference on Artificial Intelligence* (2026).
2. Crasto, R. & **Rolf, E.** Latent Domain Modeling Improves Robustness to Geographic Shifts. in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (Findings track)* (2026).
3. Rao, A., Rußwurm, M., Klemmer, K. & **Rolf, E.** Measuring the Intrinsic Dimension of Earth Representations. in *Proceedings of the International Conference on Learning Representations (ICLR)* (2026).
4. Klemmer, K., **Rolf, E.**, Robinson, C., Mackey, L. & Rußwurm, M. SatCLIP: Global, general-purpose location embeddings with satellite imagery. in *Proceedings of the AAAI Conference on Artificial Intelligence* (2025).
5. Gordon, L., **Rolf, E.** & Tambe, M. Combining diverse information for coordinated action: Stochastic bandit algorithms for heterogeneous agents. in *Proceedings of the European Conference on Artificial Intelligence (ECAI)* (2024).
6. **Rolf***, E., Klemmer, K., Robinson, C. & Kerner*, H. Position: Mission critical–Satellite data is a distinct modality in machine learning. in *Proceedings of the International Conference on Machine Learning (ICML)* (2024). ★ **Spotlight designation (3.5% acceptance rate)**.
7. Rolnick, D., Aspuru-Guzik, A., Beery, S., Dilkina, B., Donti, P. L., Ghassemi, M., Kerner, H., Monteleoni, C., **Rolf, E.**, Tambe, M., *et al.* Application-driven innovation in machine learning. in *Proceedings of the International Conference on Machine Learning (ICML)* (2024).
8. Rußwurm, M., Klemmer, K., **Rolf, E.**, Zbinden, R. & Tuia, D. Geographic location encoding with spherical harmonics and sinusoidal representation networks. in *Proceedings of the International Conference on Learning Representations (ICLR)* (2024). ★ **Spotlight designation (5% acceptance rate)**.
9. Aiken*, E., **Rolf***, E. & Blumenstock, J. E. Fairness and representation in satellite-based poverty maps: Evidence of urban-rural disparities and their impacts on downstream policy. in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)* (2023).
10. **Rolf***, E., Malkin*, N., Graikos, A., Jojic, A., Robinson, C. & Jojic, N. Resolving label uncertainty with implicit posterior models. in *Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI)* (2022). ★ **Oral Presentation (5% acceptance rate)**.
11. **Rolf, E.**, Worledge, T. T., Recht, B. & Jordan, M. Representation matters: Assessing the importance of subgroup allocations in training data. in *Proceedings of the International Conference on Machine Learning (ICML)* (2021).
12. **Rolf, E.**, Jordan, M. I. & Recht, B. Post-estimation smoothing: A simple baseline for learning with side information. in *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)* (2020).
13. **Rolf, E.**, Simchowitz, M., Dean, S., Liu, L. T., Björkegren, D., Hardt, M. & Blumenstock, J. Balancing competing objectives with noisy data: Score-based classifiers for welfare-aware machine learning. in *Proceedings of the International Conference on Machine Learning (ICML)* (2020).
14. Liu, L. T., Dean, S., **Rolf, E.**, Simchowitz, M. & Hardt, M. Delayed impact of fair machine learning. in *Proceedings of the International Conference on Machine Learning (ICML)* (2018).
★ **Best Paper Award**.

Archival Workshop Papers

1. Kaur, A., Purohit, M., Muhawenayo, G., **Rolf, E.** & Kerner, H. Pretrain Where? Investigating How Pretraining Data Diversity Impacts Geospatial Foundation Model Performance. in *2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (2026).
2. Wang, Z., Lane, K., Cai, L., Karimzadeh, M. & **Rolf, E.** A Proxy Consistency Loss for Grounded Fusion of Earth Observation and Location Encoders. in *2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (2026).
3. Nair, R., Tokas, B., Tseng, G., **Rolf, E.** & Kerner, H. Classification Drives Geographic Bias in Street Scene Segmentation. in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (2025).
4. Rao, A. & **Rolf, E.** Using Multiple Input Modalities Can Improve Data-Efficiency and OOD Generalization for ML with Satellite Imagery. *PMLR, TerraBytes Workshop at the International Conference on Machine Learning (ICML)* (2025).
★ Earlier version received **Best Student Paper Award** at the ICLR 2025 Workshop on Machine Learning for Remote Sensing.

Other Workshop Papers

1. **Rolf, E.** Evaluation challenges for geospatial ML. *Machine Learning for Remote Sensing (ML4RS) Workshop at the International Conference on Learning Representations (ICLR)* (2023).
2. **Rolf, E.**, Packer, B., Beutel, A. & Diaz, F. Striving for data-model efficiency: Identifying data externalities on group performance. *Workshop on Trustworthy and Socially Responsible Machine Learning at the Conference on Neural Information Processing Systems (NeurIPS)* (2022).
3. Soman, S., Aiken, E., **Rolf, E.** & Blumenstock, J. Can Strategic Data Collection Improve the Performance of Poverty Prediction Models? *Artificial Intelligence for Humanitarian Assistance and Disaster Response Workshop at the Conference on Neural Information Processing Systems (NeurIPS)* (2022).
4. **Rolf, E.**, Simchowitz, M., Dean, S., Liu, L. T., Björkegren, D., Hardt, M. & Blumenstock, J. Balancing Competing Objectives for Welfare-Aware Machine Learning with Imperfect Data. *AI for Social Good workshop at the Conference on Neural Information Processing Systems (NeurIPS)* (2019).
★ **Best Paper Award.**
5. Bolliger, I., Carleton, T., Hsiang, S., Kadish, J., Proctor, J., Recht, B., **Rolf, E.** & Shankar, V. Ground Control to Major Tom: the importance of field surveys in remotely sensed data analysis. *Bloomberg Data for Good Exchange* (2017).

Extended Abstracts

1. **Rolf, E.**, Klemmer, K. & Rußwurm, M. Earth Embeddings: Harnessing the Information in Earth Observation Data with Machine Learning. in *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Frontiers* (2025), 1–2.

News and Editorials

1. Klemmer, K. & **Rolf, E.** Satellite images reveal untracked human activity on the oceans. *Nature News and Views* (2024).

Preprints

1. Betti, L., Rieke, S., Obadig, I., Stewart, A. J. & **Rolf, E.** What's in an Earth Embedding? An Explainability Analysis of Location Encoders. *arXiv preprint arXiv:2606.24997* (2026).
2. Corley, I., Rao, A., **Rolf, E.**, Klemmer, K., Shelhamer, E., Lehmann, N., Rußwurm, M., Mai, G., Jacobs, N. & Kerner, H. MIND the Gap: A Geographic Implicit Neural Representation with Adjustable Spatial Scale. *arXiv preprint arXiv:2609.25454* (2026).
3. Lane, K., Wang, Z., **Rolf, E.** & Karimzadeh, M. SLED: Scalable Location Encoding via Distillation. *arXiv preprint arXiv:2608.06612* (2026).
4. Zhang, H., Betti, L., Klemmer, K., **Rolf, E.** & Alvarez-Melis, D. OT on the Map: Quantifying Domain Shifts in Geographic Space. *arXiv preprint arXiv:2604.16220* (2026).
5. Christensen, D., Carleton, T., **Rolf, E.**, Molitor, C., Biswas, S., Yan, K. & Blair, G. Estimating the Footprint of Artisanal Mining in Africa. *National Bureau of Economic Research* (2025).

White Papers

1. Elger, K., Kingsley, D., Inadomi, B., Raia, N., Rao, Y., **Rolf, E.**, Schuster, D. & Shingledecker, S. Preserving open data as a world heritage to secure our future. *Authorea Preprints* (2024).
2. Xu*, L., **Rolf*, E.**, Beery, S., Bennett, J. R., Berger-Wolf, T., Birch, T., Bondi-Kelly, E., Brashares, J., Chapman, M., Corso, A., Davies, A., Garg, N., Gaylard, A., Heilmayr, R., Kerner, H., Klemmer, K., Kumar, V., Mackey, L., Monteleoni, C., Moorcroft, P., Palmer, J., Perrault, A., Thau, D. & Tambe, M. Reflections from the Workshop on AI-Assisted Decision Making for Conservation. *arXiv preprint arXiv:2307.08774* (2023).

Invited Talks

- 2026 EarthVision workshop at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (keynote)
Foundation models beyond the visible spectrum workshop at the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (plenary)
ESIIL Innovation Summit (keynote)
Eric and Wendy Schmidt Center for Data Science & Environment at Berkeley lunch talk
Google Boulder ML chalk talk
- 2025 CU Boulder Sustainability Research Initiative AI & Data Centers Workshop (keynote)
CMCC Workshop on Artificial Intelligence for Carbon
Oregon State University AI Seminar
SIGGRAPH Frontiers talk
University of British Columbia Seminar
Maxar Technologies Seminar
CU Boulder “MathGeo” Seminar
CU Boulder Ecology and Evolutionary Biology Seminar
- 2024 NeurIPS workshop on Bayesian Decision-making and Uncertainty
NeurIPS 2024 Workshop on Tackling Climate Change with Machine Learning (keynote)
CU Boulder Earth Lab Environmental Data Science Seminar Series
Harvard Salata Institute for Climate and Sustainability Scholar Seminar Series
Harvard GeoAI Conference
- 2023 Boston University Center for Computing and Data Sciences Machine Learning Seminar
Harvard Center for Geographic Analysis Seminar Series
Google Responsible ML reading group
- 2022 Summer at Census (virtual) Workshop
Microsoft New England ML Ideas Seminar
- 2021 United States Census Bureau
Berkeley School of Information Data Science Immersion Experience
Harvard Center for Research on Computation and Society Rising Star Speaker Series
- 2020 CGIAR Big Data in Agriculture: Digital Dynamism for Adaptive Food Systems
- 2019 ImageXD Conference at Berkeley Institute of Data Science
Sigmetrics: highlights beyond Sigmetrics

Contributed Talks

- 2026 8th National Oceanic and Atmospheric Administration (NOAA) AI Workshop

Teaching

CSCI 4622: Machine Learning (Undergraduate), CU Boulder

Semesters taught: Fall 2026

Enrollments: 46 students

CSCI 5622: Machine Learning (Graduate), CU Boulder

Semesters taught: Spring 2025 & Spring 2026

Enrollments: 28-50 students

CSCI 7000: Current Topics in CS: Geospatial and Statistical ML, CU Boulder

Semesters taught: Fall 2024 & Fall 2025

Enrollments: 11-17 students

Graduate and undergraduate teaching roles:

DS 102: Data, Inference, and Decisions, UC Berkeley, Fall 2019. (graduate student instructor)

CS 189/289A: Introduction to ML, UC Berkeley, Fall 2018. (graduate student instructor)

COS 340: Reasoning about Computation, Princeton, Fall 2015, Spring 2016. (undergrad TA & grader)

Research Advising

Ph.D. Students

Livia Betti, CU Boulder Computer Science Ph.D. student, 2024-

Recipient of CU Boulder Chancellor's Fellowship and NSF GRFP Fellowship

Allison Xu, CU Boulder Computer Science Ph.D. student, 2026-

Benjamin Herrera, CU Boulder Computer Science Ph.D. student (co-advised with Mark Zhao), 2026-

Recipient of NSF GRFP Fellowship

Postdocs

Levi "VeeVee" Cai, ESIIL AI and Environment Postdoctoral Fellow, 2025-

Brian Lee, ESIIL Postdoctoral Fellow (co-mentored with Katherine Siegel), 2026-

Sam Lapp, ESIIL Postdoctoral Fellow (co-mentored with Sara Beery), 2026-

Master's students

Blake Dehaas, CU Boulder Computer Science Master's, 2026

Juliette Sikking, Visiting Master's thesis student from EPFL, 2026

Sebastian Rieke, CU Boulder Computer Science Master's, 2026

Ian Christie, CU Boulder Computer Science Master's, 2025-2026

Lucia Witikko, CU Boulder Computer Science Master's, 2025-2026

Undergraduate Students

Shona Doyle, CU Boulder Computer Science Undergraduate, 2026

Pranav Konijeti, CU Boulder Computer Science Undergraduate, 2026

Theodora Worledge, UC Berkeley Computer Science Undergraduate, 2019-2021

→ *Ph.D. student at Stanford Computer Science, NSF GRFP recipient*

Professional Service

Advisory Roles

Steering Committee, Clay v2 foundation 2026-

- Guide open-source geospatial AI efforts of the Clay v2 foundation

American Geophysical Union (AGU) Data Position Statement Panel, 2023-2024

- Updated the Data Position Statement of the American Geophysical Union.

Academic and Professional Panels

2026 CU Boulder Sustainability Research Initiative Summit

2024 SatCamp Geospatial Event Conference

CU Boulder AI Community of Practice AI Showcase

Computer Vision and Pattern Recognition (CVPR) EarthVision Workshop

Data.org Accelerate Conference

Area Chair

Conferences NeurIPS Datasets and Benchmarks track

Program Committee/Reviewer

Journals Science, Environmental Data Science, IEEE Robotics and Automation Letters

Conferences NeurIPS, ICML, ICLR, AAAI, AAAI AI for Social Impact track, AISTATS, EAAMO, UAI

Workshops ML for Remote Sensing, NeurIPS AI + Humanitarian Assistance and Disaster Response, Consequential Decision Making in Dynamic Environments, Climate Change AI

Workshop Organization

2026 International Society for Photogrammetry and Remote Sensing (ISPRS) Thematic Session on "Earth embeddings"

2026 ICLR Workshop on Machine Learning for Remote Sensing, lead organizer

2025 ICLR Workshop on Machine Learning for Remote Sensing

2022 Harvard CRCS Workshop on AI-Assisted Decision-Making for Conservation

2021 NeurIPS Workshop on AI for Humanitarian Assistance and Disaster Response

2020 Good Data Seminar (UC Berkeley cross-disciplinary research seminar)

2019 Good Data Seminar (UC Berkeley cross-disciplinary research seminar)

2018 Good Data Seminar (UC Berkeley cross-disciplinary research seminar)

Departmental Service

Executive Committee, tenure-track faculty representative (elected position) 2026-

Graduate Committee, subcommittee on curriculum and degree requirements 2025-2026

Graduate Committee, subcommittee on student mentorship and awards 2024-2025

Industry Research Internships

Research Intern, Google Research, November 2021 - January 2022

Research Intern, Microsoft Research, May 2021 - July 2021

Last updated: October 6, 2026